



SkillVane IT Academy
SKILLS THAT SHAPE CAREERS

2026 CURRICULUM

PROFESSIONAL CAREER PROGRAM

GCP Data Engineering + AI Fundamentals

Industry-Focused Training with Real-World Projects

• **STRUCTURED LEARNING** • **HANDS-ON LABS** • **INDUSTRY PROJECTS**



COURSE TRAINER

Shaik Saidhul

Data Engineering Instructor

Course Snapshot

EVERYTHING AT A GLANCE



ROLE
GCP Data Engineer



COURSE DURATION
3 months



MODE
Online



LANGUAGE
English



Your Learning Journey

A structured path from cloud foundations to production-ready data engineering and AI skills.

01

GCP Cloud Basics

GCP Introduction • GCP Interfaces • GCP Locations • GCP IAM & Admin • +1 more

02

Python for Data Engineer

Python Learning Path

03

GCP Data Engineering Tools

Google Cloud Storage • Cloud SQL • BigQuery (SQL development) • DataProc (Pyspark Development) • +7 more

04

AI Fundamentals for GCP Data Engineers (Current Market Demand)

AI & Generative AI Learning Path

05

Complementary End-to-End Projects:

Included End-to-End Projects

06

By the End of the course What Students can Expect

Proficient in SQL Development • Pyspark Development Skills • Apache Beam Development Mastery • DAG Creations with Cloud Composer • +4 more

07

Course Highlights:

What Is Included

GCP Cloud Basics

5 FOCUSED TOPICS

01

GCP Introduction

- The need for cloud computing in modern businesses.
- Key features and offerings of Google Cloud Platform (GCP).
- Overview of core GCP services and products.
- Benefits and advantages of using cloud infrastructure.
- Step-by-step guide to creating a free-tier account on GCP.

02

GCP Interfaces

Console

- Navigating the GCP Console
- Configuring the GCP Console for Efficiency
- Using the GCP Console for Service Management

Shell

- Introduction to GCP Shell
- Command-line Interface (CLI) Basics
- GCP Shell Commands for Service Deployment and Management

SDK

- Overview of GCP Software Development Kits (SDKs)
- Installing and Configuring SDKs
- Writing and Executing GCP SDK Commands

03

GCP Locations

Regions

- Understanding GCP Regions
- Selecting Regions for Service Deployment
- Impact of Region on Service Performance

Zones

- Exploring GCP Zones
- Distributing Resources Across Zones
- High Availability and Disaster Recovery Considerations

Importance

- Significance of Choosing the Right Location
- Global vs. Regional Resources
- Factors Influencing Location Decisions

Identities

- Introduction to Identity and Access Management (IAM)
- Users, Groups, and Service Accounts
- Best Practices for Identity Management

Roles

- GCP IAM Roles Overview
- Defining Custom Roles
- Role-Based Access Control (RBAC) Implementation

Policy

- Resource-based Policies
- Understanding and Implementing Organization Policies
- Auditing and Monitoring Policies

Resource Hierarchy

- GCP Resource Hierarchy Structure
- Managing Resources in a Hierarchy
- Organizational Structure Best Practices

- Getting started with Linux
- Linux Installation

Basic Linux Commands

- Cloud shell tips
- File and Directory Operations (ls, cd, pwd, mkdir, rmdir, cp, mv, touch, rm, nano)
- File Content Manipulation (cat, less, head, tail, grep)
- Text Processing (awk, sed, cut, sort, uniq)
- User and Permission related (whoami, id, su, sudo, chmod, chown)

Python for Data Engineer

1 FOCUSED TOPIC

01

Python Learning Path

- Getting Started with Python
- Introduction to Python and basic functions

Data Types

- Strings
- Operators
- Numbers (Int, Float)
- Booleans

Data Structures

- Lists
- Tuples
- Dictionaries
- Sets

Python Programming Constructs

- if, elif, else statements
- for loops, while loops

File I/O operations

- What is FILE IO operations
- Read, write, append, rename, remove, copy, move

Exception Handling and Logging

- what is exception handling, why it is needed
- try and except blocks
- else, finally, raise blocks
- Why Logging?
- Logging Levels
- Writing Logs to File
- Logging Best Practices

Data and Time operations

- Working date and time
- Formatting
- Parsing
- Time delta

Modular Programming in Python

- Functions & Lambda Functions
- Classes

REST API basics

- What is API
- HTTP methods (GET, POST, PUT, DELETE)
- Status Codes, Request Headers, JSON Responses
- Using the requests Library, Reading API Responses

Modules and Packages

- What is module and packages
- Importing
- Built-in modules
- Install packages

GCP Data Engineering Tools

11 FOCUSED TOPICS

01

Google Cloud Storage

- Overview of Cloud Storage as a scalable and durable object storage service.
- Understanding buckets and objects in Cloud Storage.
- Use cases for Cloud Storage, such as data backup, multimedia storage, and website content

Labs: using console & CLI to do below

- Creating and managing Cloud Storage buckets.
- Uploading and downloading objects to and from Cloud Storage.
- Setting access controls and permissions for buckets and objects.
- Data Transfer and Lifecycle Management
- Versioning and Object Versioning

- Integration with Other GCP Services
- Monitoring and logging for Cloud Storage operations.

02

Cloud SQL

- Introduction to Cloud SQL
- Creating and Managing Cloud SQL Instances
- Configuring database settings, users, and access controls.
- Connecting to Cloud SQL instances using Cloud SQL studio, Shell, Workbenches
- Importing and exporting data in Cloud SQL.
- Backups and High Availability
- Integration with Other GCP Services
- Managing database user roles and permissions.
- Introduction to DMS

End to End Database migration Project

- Manual: Export and Import method
- Automation: Cloud SQL DMS method

- Introduction to BigQuery
- BigQuery Architecture
- Use cases for BigQuery in business intelligence and analytics.
- Various method of creating table in BigQuery
- BigQuery Data Sources and File Formats
- Native table and External Tables

Working with complex data types

- Working json data, nested, repeated and array data

Data Integration and Export

- Loading data into BigQuery from Cloud Storage, Cloud SQL, and other sources.
- Exporting data from BigQuery to various formats.
- Real-time data streaming into BigQuery.

- Configuring access controls and permissions in BigQuery.

BigQuery Views:

- Views
- Materialized Views
- Authorized Views

- Optimization techniques in BigQuery
- BigQuery Slots - on demand, flat-rate, flex-slots
- Case Study-1: implement a real-world analytics data platform for Spotify
- Case Study-2: Enterprise Social Media analytics platform

- Introduction to Hadoop and Apache Spark
- Understanding the difference between Spark and MapReduce
- What is Spark and Pyspark.
- Understanding Spark framework and its functionalities
- Overview of DataProc as a fully managed Apache Spark and Hadoop service.

Cluster Creation and Configuration

- Creating and managing DataProc clusters.
- Configuring cluster properties for performance and scalability.
- Preemptible instances and cost optimization.

Learning Pyspark:

- How to read from multiple data sources - csv, text, json, parquet, database table, BigQuery tables
- How to perform multiple transformations
- How to write to multiple targets - csv, text, json, parquet, database table, BigQuery tables

Running Jobs on DataProc

- Submitting and monitoring Spark and Hadoop jobs on DataProc.
 - Use of initialization actions and custom scripts.
 - Job debugging and troubleshooting.
-
- Case study-1: Data Cleaning of Employee Travel Records
 - Case study-2: Processing real-time patient health data
 - Case study-3: Creating a pyspark job to support ML model creations

05

DataFlow (Apache Beam development)

- Introduction to DataFlow
- Use cases for DataFlow in real-time analytics and ETL.
- Understanding the difference between Apache Spark and Apache Beam
- How Dataflow is different from Dataproc

Learning Apache Beam

- How to read from multiple data sources - csv, text, json, parquet, database table, BigQuery tables
 - How to perform multiple transformations
 - How to write to multiple targets - csv, text, json, parquet, database table, BigQuery tables
-
- Case study-1: Template method of creating pipelines
 - Case study-2: E-commerce Transaction Processing
 - Case study-3: End to End Streaming Pipeline using Apache beam with Dataflow, Python app, PubSub, BigQuery, GCS

06

Cloud Pub/Sub (Streaming)

- Introduction to Pub/Sub
- Understanding the role of Pub/Sub in event-driven architectures.
- Key Pub/Sub concepts: topics, subscriptions, messages, and acknowledgments.

Creating and Managing Topics and Subscriptions

- Using the GCP Console to create Pub/Sub topics and subscriptions.
- Configuring message retention policies and acknowledgment settings.

Publishing and Consuming Messages

- Writing and deploying code to publish messages to a topic.
- Implementing subscribers to consume and process messages from subscriptions.

- Case study-1: Streaming use-case using Dataflow

07

Cloud Composer (DAG Development)

- Introduction to Composer/Airflow
- Overview of Airflow Architecture
- Use cases for Composer in managing and scheduling workflows.

Creating and Managing Workflows

- Creating and configuring Composer environments.
- Defining and scheduling workflows using Apache Airflow.
- Monitoring and managing workflow executions.

Integration with Data Engineering Services

- Orchestrating workflows involving BigQuery, DataFlow, and other services.
- Coordinating ETL processes with Composer.
- Integrating with external systems and APIs.

Error Handling and Troubleshooting

- Handling errors and retries in Composer workflows.
- Debugging and troubleshooting failed workflow executions.
- Logging and monitoring for Composer workflows.

- Level-1-DAG: Orchestrating the BigQuery pipelines
- Level-2-DAG: Orchestrating the DataProc pipelines
- Level-3-DAG: Orchestrating the Dataflow pipelines
- Deploy DAGs: Implementing CI/CD in Composer Using Cloud Build and GitHub

08

Databricks on GCP

- What is Lakehouse Architecture?

Difference Between:

- Data Lake
- Data Warehouse
- Data Lakehouse

Introduction to Databricks

- Overview of Databricks
- Databricks Architecture
- Setting up a Databricks Workspace

Unity Catalog - Unified Data Governance

- Introduction to Unity Catalog
- Core Concepts: Metastore, Catalog, Schema, Tables, Volumes, Functions, Models
- External Data Access
 - Storage Credentials
 - External Locations (S3, GCS, ADLS)
- Lakehouse Federation - Foreign Catalog, Connection
- Delta Sharing - Share, Recipient, Provider

Databricks Clusters

- Introduction to Clusters
- Types of Clusters
- Cluster Modes

- DBUtils Commands - File handling, Notebook Widgets, Secrets

Introduction to Delta Lake

- What is Delta Lake?
- Creating Delta Lake Tables - Managed Tables, External Tables
- Understanding Delta Lake Table Creation - Delta Log, Parquet Data Files
- Advanced Delta Lake Features
 - Versioning
 - Time Travel
 - Compacting
 - Liquid clustering
 - Vacuum

Databricks Jobs & Pipelines

- Creating a Job
- Scheduling Jobs
- Setting Parameters
- Managing Dependencies
- Setting up Alerts

- Case study-1: Structured Streaming with Databricks
- Case study-2: Incremental Data Loading with Auto Loader

09

Data Fusion (complementary)

Introduction to Data Fusion

- Overview of Data Fusion as a fully managed data integration service.
- Use cases for Data Fusion in ETL and data migration.

Building Data Integration Pipelines

- Creating ETL pipelines using the visual interface.
- Configuring data sources, transformations, and sinks.
- Using pre-built templates for common integration scenarios.

Integration with GCP and External Services

- Integrating Data Fusion with BigQuery, Cloud Storage, and other GCP services.

- Case study-1: End to End pipeline using Data fusion with Wrangler, GCS, BigQuery

10

Cloud Functions (complementary)

- Cloud Functions Introduction
- Setting up Cloud Functions in GCP
- Event-driven architecture and use cases
- Writing and deploying Cloud Functions

Triggering Cloud Functions:

- HTTP triggers
- Pub/Sub triggers
- Cloud Storage triggers

- Monitoring and logging Cloud Functions
- Usecase-1: Loading the files from GCS to BigQuery as soon as it is uploaded.

11

Terraform (complementary)

- Terraform Introduction
- Installing and configuring Terraform.
- Infrastructure Provisioning

Terraform basic commands

- Init, plan, apply, destroy

Labs: Create Resources in Google Cloud Platform

- GCS buckets
- Dataproc cluster
- BigQuery Datasets and tables
- And more resources as needed

MODULE 04

AI Fundamentals for GCP Data Engineers (Current Market Demand)

1 FOCUSED TOPIC

01

AI & Generative AI Learning Path

Introduction to Artificial Intelligence & Generative AI

- Understanding AI, ML, Deep Learning, Generative AI, and LLMs.
- Real-world AI use cases in Data Engineering and Analytics.

Machine Learning Fundamentals

- Supervised, Unsupervised, and Reinforcement Learning.
- Training, validation, testing, and model evaluation concepts.

Prompt Engineering for Data Engineers

- Designing effective prompts for Gemini, ChatGPT, and Vertex AI.
- Using AI for SQL generation, code generation, documentation, and data analysis.

Google Vertex AI Fundamentals

- Overview of Vertex AI ecosystem.
- Working with foundation models, Model Garden, and Gemini APIs.

AI-Powered Data Pipelines

- Integrating AI services with BigQuery, GCS, Dataflow, and Dataproc.
- Building intelligent ETL/ELT pipelines using AI-driven transformations.

MODULE 05

Complementary End-to-End Projects:

1 FOCUSED TOPIC

01

Included End-to-End Projects

- Healthcare project on GCP of 8+ hours
- Road Traffic project on Databricks of 5+ hours

MODULE 06

By the End of the course What Students can Expect

8 FOCUSED TOPICS

01

Proficient in SQL Development

- Mastering SQL for querying and manipulating data within Google BigQuery and Cloud SQL.
- Writing complex queries and optimizing performance for large-scale datasets.
- Understanding schema design and best practices for efficient data storage.

02

Pyspark Development Skills

- Proficiency in using PySpark for large-scale data processing on Google Cloud.
- Developing and optimizing Spark jobs for distributed data processing.
- Understanding Spark's RDDs, Dataframes, and transformations for data manipulation.

03

Apache Beam Development Mastery

- Creating data processing pipelines using Apache Beam.
- Understanding the concepts of parallel processing and data parallelism.
- Implementing transformations and integrating with other GCP services.

04

DAG Creations with Cloud Composer

- Designing and implementing Directed Acyclic Graphs (DAGs) for orchestrating workflows.
- Using Cloud Composer for workflow automation and managing dependencies.
- Developing DAGs that integrate various GCP services for end-to-end data processing.

05

Notebooks, Workflows with Databricks

- Understand how to build and manage data pipelines using Databricks and Delta Lake.
- Efficiently query and analyze large datasets with Databricks SQL and Apache Spark.
- Implement scalable workflows and optimize performance within Databricks.

06

Architecture Planning

- Proficient in architecting end-to-end data solutions on GCP.
- Understanding the principles of designing scalable, reliable, and cost-effective data architectures.

07

Certification Readiness

- Prepare for the Google Cloud Professional Data Engineer (PDE) and
- Associate Cloud Engineer (ACE) certifications through a combination of theoretical knowledge and hands-on experience.

08

ML & AI ready

- Students will understand how modern AI and Generative AI integrate with Google Cloud Data Engineering solutions and will be able to build AI-enabled data platforms using BigQuery, Vertex AI, and Gemini.

Course Highlights:

1 FOCUSED TOPIC

01

What Is Included

- Hands-on Labs & Real-world Use Cases
- Live Q&A Sessions
- Course Completion Certificate
- Access to Study Materials & Code Repository
- Industry-Standard Best Practices

“

The course will empower students with practical skills in SQL, PySpark, Apache Beam, DAG creations, and architecture planning, ensuring they are well-prepared to tackle real-world data engineering challenges and successfully obtain GCP certifications.

Thank You.

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